

Iram Azam

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PROFESSIONAL SUMMARY

Software and Research Engineer with hands-on experience in applied AI, high-performance data analytics, and scalable ML pipelines. Three peer-reviewed publications in applied ML and AI across biomedical NLP and health data classification. Proven ability to deliver advanced technical training, collaborate with multidisciplinary teams, and communicate complex findings to diverse audiences.

EDUCATION

Master of Science in Technology — Concentration: Computer Information Technology May 2026
Purdue University | GPA: 4.00/4.00

Bachelor of Science in Computer Science January 2022
COMSATS Institute of Information Technology | GPA: 3.35/4.00

PEER-REVIEWED PUBLICATIONS

EMG-Based Rehabilitation Dosage Optimization Using Machine Learning — *Bioengineering*, 2026 ([Link](#))

- Built a constrained optimization model in Python to personalize 60-minute physical therapy sessions across 4 rotator cuff exercises
- Applied XGBoost to predict shoulder muscle activation from sEMG signals with $r = 0.73$ Pearson correlation across unseen subjects
- Outperformed 3 competing algorithms (SVR, KNN, AdaBoost) in cross-subject EMG prediction while reducing model training time to 43 seconds

Biomedical Concept Normalization Using Neural Embeddings and LLMs — *EACL* 2026 ([Link](#))

- Designed a SapBERT + FAISS pipeline that normalized health concepts across 3.59 million UMLS embeddings with 85.8% accuracy on clinical EHR data
- Processed and analyzed large healthcare-related datasets including 3,000+ clinical notes and 600+ public health-related social media records, improving accuracy from 23.5% to 98.0% using GPT-4o mini
- Benchmarked 7 LLMs (GPT-4o, GPT-5, Gemini, Llama) for health concept normalization, integrating generative and embedding-based methods into a unified 2-stage pipeline

Batch Inference Pipeline for Health Text Classification Using LLMs — *IEEE Big Data* 2025 ([Link](#))

- Developed an LLM-based batch classification method that processed 12,331 health-related tweets using GPT-4o, Gemini, and GPT-4o Mini across batch sizes up to 1,000
- Identified Gemini 2.0 Flash as the most cost-effective model at optimal batch size of 150, classifying 12,331 tweets for just \$0.005 and outperforming the word embedding + LSTM baseline in F1 score
- Benchmarked 4 LLMs across 13 batch sizes on throughput, latency, accuracy, and F1, showing GPT-4o achieved the best classification accuracy (84.3%) at batch size 50
- Applied prompt engineering, batch prompting, and error analysis to improve reliability and support publication-ready computational results

EXPERIENCE

Teaching Assistant | Purdue University Aug 2024 – Present

- Developed and delivered lectures and labs in software design and debugging for 200+ students
- Evaluated student code quality, provided technical feedback, and mentored students on readable, maintainable, and well-documented programming practices
- Collaborated with faculty to design curriculum and lab materials aligned with industry-standard software development

Software Engineer | Pattern Technologies Jan 2024 – Jul 2024

- Developed and shipped consumer-facing React Native mobile app features for online ordering and loyalty platform, supporting 50+ cloud kitchen locations
- Revamped UI/UX and optimized cross-platform mobile features for iOS and Android, enhancing CRM workflows, contributing to a client achieving 700% revenue growth within 6 months of launch

- Translated complex technical findings into clear reports for cross-functional stakeholders, streamlining delivery of core app features

Software Engineer | Dastgyr Technologies

Jan 2022 – Dec 2023

- Built scalable React and Node.js web portals on the admin side for a \$37M Series A-funded global B2B trade marketplace serving 100k+ SMBs
- Integrated REST APIs using JavaScript to enable real-time order monitoring, user management, and reporting workflows across 4 admin portals supporting 1.5M+ orders
- Collaborated with product managers, designers, and engineers in Agile Scrum sprints, managing end-to-end feature delivery via Jira from PRD and Figma review through coding, Jest unit testing, and production debugging

PROJECTS

AI-Based Fitness and Nutrition Assistant Using Multi-Agent System

Spring 2026

- Designed a multi-agent AI system using Python, LangChain, and Google Gemini API with 3 specialized agents to generate personalized fitness and nutrition recommendations
- Implemented RAG pipeline using FAISS vector database and HuggingFace embeddings to process fitness and nutrition PDFs for similarity-based knowledge retrieval, improving response accuracy
- Integrated 3 external APIs for BMI/calorie calculation, memory handling, and safety filtering for structured plan generation tailored to user goals

Heart Disease Mortality Rate Analysis Using Statistical Methods

Fall 2024

- Analyzed gender disparities in heart disease mortality rates among 353 Illinois residents aged 35-64 using publicly available government data
- Applied statistical methods including Chi-Square testing, One-Way ANOVA, descriptive statistics, and variance analysis to identify significant healthcare patterns
- Created data visualizations using charts and summarized findings through descriptive statistics and variance analysis to support evidence-based public health insights and recommendations

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, HTML5, CSS3, SQL, C, C++

Frameworks: React, React Native, Node.js, Express.js, LangChain, Bootstrap, Material UI

Databases & Tools: MySQL, PostgreSQL, MongoDB, Firebase, Git, GitLab, Jest, Jira

AI/ML & Data Science: LLMs, NLP, Embeddings, Scikit-learn, TensorFlow, PyTorch, Keras, HuggingFace, Pandas, NumPy, Matplotlib

Design & Testing: UI/UX Design Principles, Responsive Design, System Design, Unit & Integration Testing

GenAI: Anthropic Claude, OpenAI GPT, Google Gemini, Meta Llama, Agentic AI

AWARDS & LEADERSHIP

Vice President, Women in IT (WIIIT) — Purdue University

Spring 2025 – Present

- Lead professional development and student engagement initiatives promoting diversity and inclusion in technology

Graduate Research Award — Purdue University

Fall 2025

- Awarded for excellence in graduate-level research contributions and academic performance in technology and artificial intelligence research

Campus Ambassador — Purdue University

Fall 2025

- Represented the university through student outreach, campus engagement, and promotional activities

Student Council Leadership Scholarship — Purdue University

Fall 2025

- Recognized for outstanding leadership and active involvement in student and campus initiatives

College of Technology Scholarship — Purdue University

Spring 2025

- Received a merit-based scholarship in recognition of academic excellence, leadership, and commitment to technology